

UD65640

Application Note

Final Test for FPC only TE pad

AN-K0017-250527

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## 1. Target

Test UD65640 CMOS final test, check IC Status is normal.

### Test List

| No | Item               | Description                                                      | Use Pad Name |
|----|--------------------|------------------------------------------------------------------|--------------|
| 1  | OSC Frequency Trim | Trim TE single at 60Hz.                                          | TE           |
| 2  | Display BIST Test  | Check whether the display active mode is normal                  |              |
| 3  | OTP Program        | Programmer trim value and check whether the reload OTP is normal |              |
|    |                    |                                                                  |              |

Table 1

## 2. Register

### Access Control: Manufacturer Command Access Control (80h)

| 80h          | Access Control                                                                                                                                                                                                   |    |    |    |    |    |    |    |        |     |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----|----|--------|-----|
|              | W/R                                                                                                                                                                                                              | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0     | Hex |
| Command      | —                                                                                                                                                                                                                | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0      | 80h |
| Parameter 1  | W/R                                                                                                                                                                                                              | 0  | 0  | 0  | 0  | 0  | 0  | 0  | MCS_EN | 00h |
| Descriptions | MCS_EN:<br>This command releases parameters so that Manufacturer Command Access are enabled.<br>1: Access enabled.<br>0: Access denied. (Protection on)<br><br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |    |    |    |    |    |    |    |        |     |
| Restrictions |                                                                                                                                                                                                                  |    |    |    |    |    |    |    |        |     |

### Device Code: Device Code (88h)

| 88h          | Device Code                                                                                                                                                                                           |           |    |    |    |    |    |    |    |     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----|----|----|----|----|----|----|-----|
|              | W/R                                                                                                                                                                                                   | B7        | B6 | B5 | B4 | B3 | B2 | B1 | B0 | Hex |
| Command      | —                                                                                                                                                                                                     | 1         | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 88h |
| Parameter 1  | W/R                                                                                                                                                                                                   | DDB0[7:0] |    |    |    |    |    |    |    | 00h |
| Parameter 2  | W/R                                                                                                                                                                                                   | DDB1[7:0] |    |    |    |    |    |    |    | 00h |
| Parameter 3  | W/R                                                                                                                                                                                                   | DDB2[7:0] |    |    |    |    |    |    |    | 00h |
| Descriptions | DDB0:<br>This command is for user command (DAh) version number.<br>DDB1:<br>This command is for user command (DBh) version number.<br>DDB2:<br>This command is for user command (DCh) version number. |           |    |    |    |    |    |    |    |     |
|              | 1 = Must fill 1; 0 = Must fill 0; X = Don't care.                                                                                                                                                     |           |    |    |    |    |    |    |    |     |
| Restrictions |                                                                                                                                                                                                       |           |    |    |    |    |    |    |    |     |

## Analog Settings: Analog Settings (90h)

| 90h          | Analog Setting                                    |                    |                   |                  |    |                     |                    |    |    |     |
|--------------|---------------------------------------------------|--------------------|-------------------|------------------|----|---------------------|--------------------|----|----|-----|
|              | W/R                                               | B7                 | B6                | B5               | B4 | B3                  | B2                 | B1 | B0 | Hex |
| Command      | —                                                 | 1                  | 0                 | 0                | 1  | 0                   | 0                  | 0  | 0  | 90h |
| Parameter 1  | W/R                                               | DV_VREFP_OPT [3:0] |                   |                  |    | 0                   | DV_TRIM_DVDD [2:0] |    |    | 70h |
| Parameter 2  | W/R                                               | 0                  | DV_TRIM_OSC [6:0] |                  |    |                     |                    |    |    | 40h |
| Parameter 3  | W/R                                               | 0                  |                   | DV_BG_TRIM [5:0] |    |                     |                    |    |    | 00h |
| Parameter 4  | W/R                                               | 0                  |                   |                  |    | DV_BG_TRIM_TC [3:0] |                    |    |    | 00h |
| Descriptions | DV_VREFP_OPT [3:0]: VREFP Voltage Output Set.     |                    |                   |                  |    |                     |                    |    |    |     |
|              | DV_TRIM_DVDD [2:0]: Trim DVDD Voltage.            |                    |                   |                  |    |                     |                    |    |    |     |
|              | DV_TRIM_OSC [6:0]: Trim OSC Frequency.            |                    |                   |                  |    |                     |                    |    |    |     |
|              | DV_BG_TRIM [5:0]: Trim BG Voltage.                |                    |                   |                  |    |                     |                    |    |    |     |
|              | 1 = Must fill 1; 0 = Must fill 0; X = Don't care. |                    |                   |                  |    |                     |                    |    |    |     |
| Restrictions |                                                   |                    |                   |                  |    |                     |                    |    |    |     |

## BIST\_CTRL: (F0h)

| F0h          | BIST_CTRL                                                                                                                                                                         |    |    |    |    |    |          |    |         |     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|----|----|----|----------|----|---------|-----|
|              | W/R                                                                                                                                                                               | B7 | B6 | B5 | B4 | B3 | B2       | B1 | B0      | Hex |
| Command      | —                                                                                                                                                                                 | 1  | 1  | 1  | 1  | 0  | 0        | 0  | 0       | F0h |
| Parameter 1  | W/R                                                                                                                                                                               | 0  |    |    |    | 0  |          |    | BIST_EN | 00h |
| Parameter 2  | W/R                                                                                                                                                                               | 0  |    |    |    |    | AGING_EN | 0  |         | 00h |
| Descriptions | BIST_EN: Build in self-test function, 1: enable / 0: disable.<br>AGING_EN : AGING Mode function, 1: enable / 0: disable.<br><br>1 = Must fill 1; 0 = Must fill 0; X = Don't care. |    |    |    |    |    |          |    |         |     |
| Restrictions |                                                                                                                                                                                   |    |    |    |    |    |          |    |         |     |

## 3. Final Test Flow:

### 3.1 OSC Frequency Trim

Step1. Register write 11h

Step2. Register write 29h

Step3. Register write 80h, 01h

Step4. Register write F0h, 01h

Step5. Register write 35h, 00h

Step6. Measure TE pin, frame rate target 60Hz.

Step7. Register adjustment 90h.P2 DV\_TRIM\_OSC [6:0] b'0000000 ~ b'1111111.

Step8. Loop Step7 TE single into find closest 60Hz and record DV\_TRIM\_OSC value to file.

## 3.2 Display BIST Test

Step1. Register write F0h,01h

Step2. Check panel display patten.

Step3. Check Step2 if display abnormal, stop and mark fail number.

## 3.3 OTP Flow

Step1. [Save all result to file.](#)

Step2. Register write F0h,00h,00h

Step3. Register write 28h.

Step4. Register write 10h.

Step5. Read register 0Ah 1Byte, check value(08h) on sleep in status.

Step6. Start OTP Flow: VPP Power on 7.5V

Step7. Register write 85h, 03h and wait >300us.

Step8. Register write 82h, 02h, 00h, 00h, 10h //OTP Program Group 2

Step9. Wait 10ms.

Step10. Register read 83h 2Byte, if read out value 00h,08h go to Step11 than go to Step9.

Step11. Register write 85h, 00h.

Step12. Register power off.

## 3.4 Re-Power on and OTP Reload

Step1. All power on.

Step2. Pin Reset

Step3. Register read 83h 5Byte if P2 G2\_WC[1:0] of them is 0 abnormal.

Step4. Register Read 90h 3Byte if value is not default value, OTP reload finish.

## 4. Revision History

| Version No.     | Date      | Page | Description     |
|-----------------|-----------|------|-----------------|
| AN-K0017-250527 | 2025/5/27 | 5    | Initial Version |